

Chapter 5: Data Analysis, 2017 data, Total Number x Total Area

Import Libraries

```
In [1]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib as mpl
from matplotlib import pyplot as plt

%matplotlib inline
```

Import the data

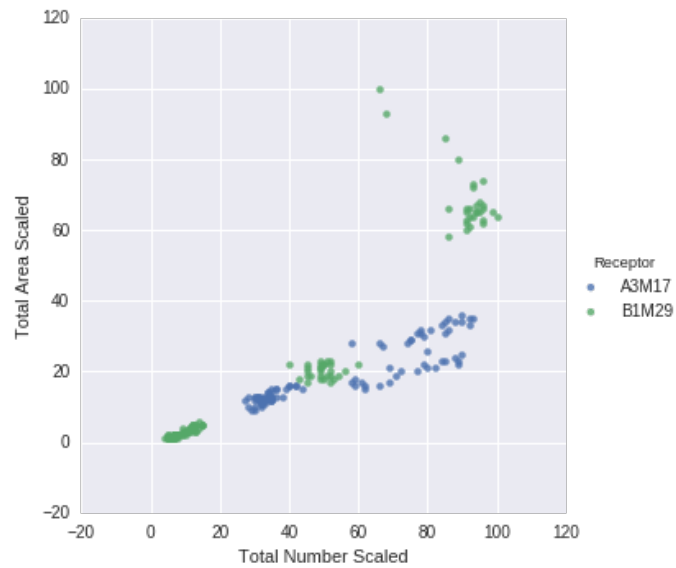
```
In [2]: totalsXX = None
totalsXX = pd.read_csv('TXX_Totals.csv')
```

Visualizations

Unless otherwise noted, the graphs will use the scaled values.

Number by area, both receptors, post steps, both runs

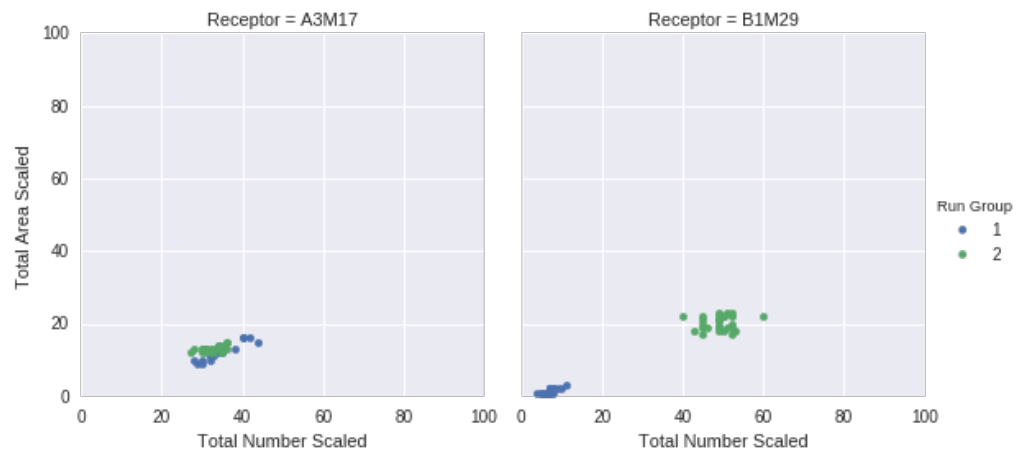
```
In [3]: number_by_area_all = sns.lmplot('Total Number Scaled', 'Total Area Scaled',  
                                         data=totalsXX,  
                                         fit_reg=False,  
                                         hue='Receptor')  
number_by_area_all.savefig('xx_number_by_area_both.png')
```



Number by area, both receptors, -nl-post

```
In [4]: tmpdf = totalsXX.loc[totalsXX['Experiment Step'] == '-nl-post']

g = sns.FacetGrid(tmpdf,
                  col='Receptor',
                  col_order=['A3M17', 'B1M29'],
                  col_wrap=2,
                  hue='Run Group',
                  size=4,
                  ylim=(0,100),
                  xlim=(0,100)
                  )
g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('xx_num-by-area_both-receptors_noligand-post.png')
```



Number by area, both receptors, -l-post

```
In [5]: tmpdf = totalsXX.loc[totalsXX['Experiment Step'] == '-1-post']

g = sns.FacetGrid(tmpdf,
                  col='Receptor',
                  col_order=['A3M17', 'B1M29'],
                  hue='Run Group',
                  col_wrap=2,
                  size=4,
                  ylim=(0,100),
                  xlim=(0,100)
                  )
g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('xx_num-by-area_both-receptors_ligand-post.png')
```



Number by area, A3M17, post steps, run group 1

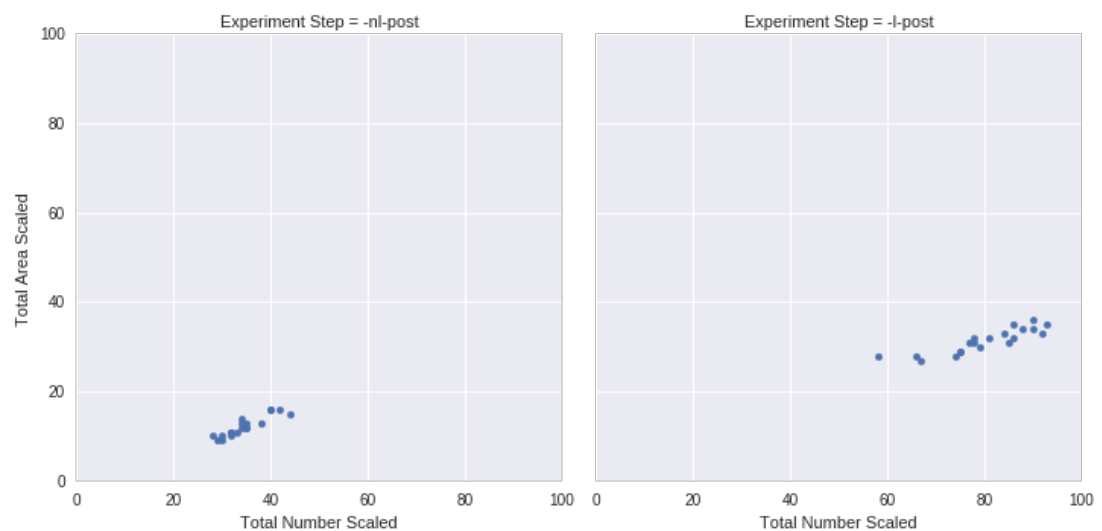
```

In [6]: tmpdf = None
tmpdf = totalsXX.loc[(totalsXX['Receptor'] == 'A3M17') &
                    (totalsXX['Experiment Step'].str.contains('post')) &
                    (totalsXX['Run Group'] == 1)
                    ]

g = sns.FacetGrid(tmpdf,
                  col='Experiment Step',
                  col_order=['-nl-post', '-l-post'],
                  col_wrap=2,
                  size=5,
                  ylim=(0,100), xlim=(0,100),
                  aspect=1)

g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('xx_num-by-area_a3m17_post-steps_run-group-1.png')

```

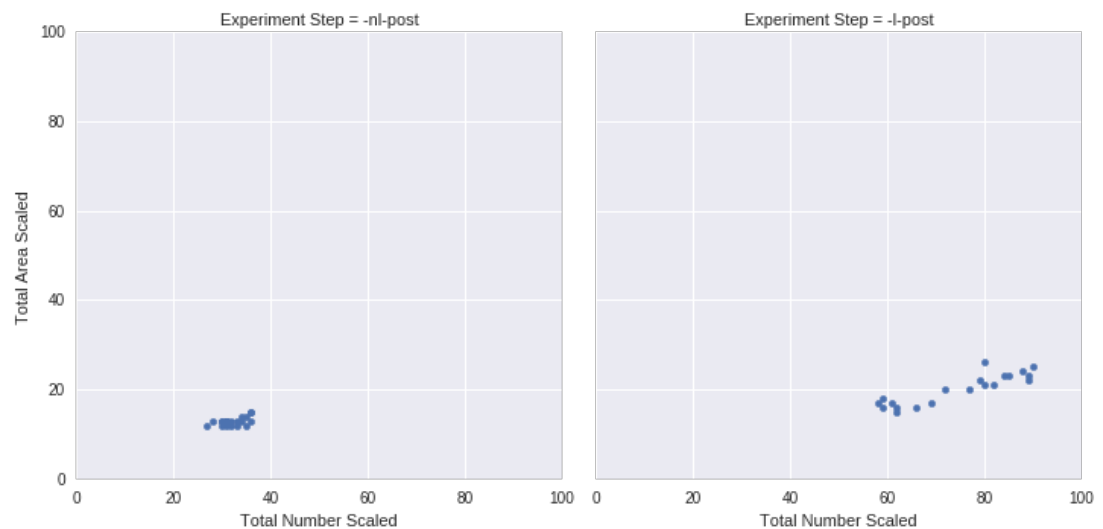


Number by area, A3M17, post steps, run group 2

```
In [7]: tmpdf = None
tmpdf = totalsXX.loc[(totalsXX['Receptor'] == 'A3M17') &
                    (totalsXX['Experiment Step'].str.contains('post')) &
                    (totalsXX['Run Group'] == 2)
                    ]

g = sns.FacetGrid(tmpdf,
                  col='Experiment Step',
                  col_order=['-nl-post', '-l-post'],
                  col_wrap=2,
                  size=5,
                  ylim=(0,100), xlim=(0,100),
                  aspect=1)

g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('xx_num-by-area_a3m17_post-steps_run-group-2.png')
```



Number by Area, A3M17, post steps, both run groups

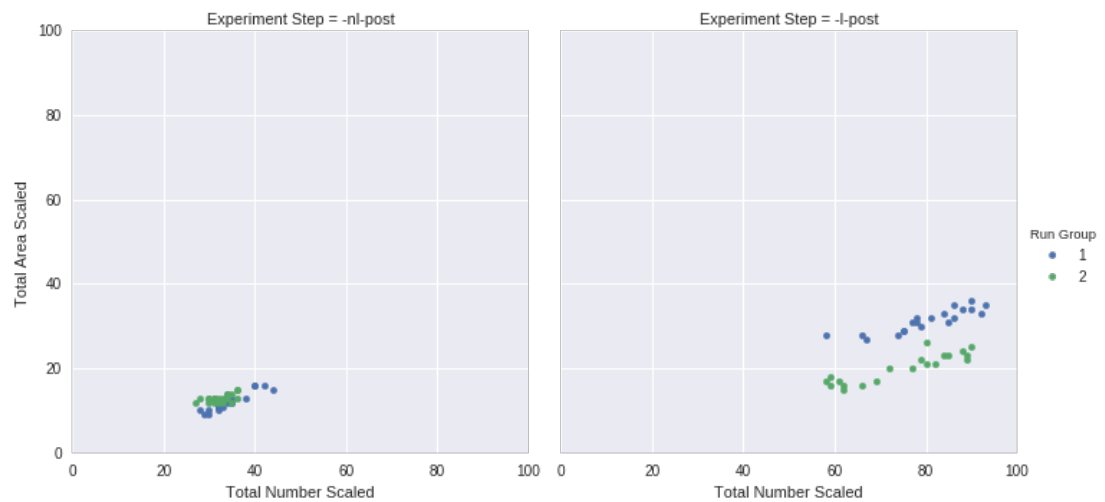
```

In [8]: tmpdf = None
tmpdf = totalsXX.loc[(totalsXX['Receptor'] == 'A3M17') &
                    (totalsXX['Experiment Step'].str.contains('post'))
                    ]

g = sns.FacetGrid(tmpdf,
                  col='Experiment Step',
                  col_order=['-nl-post', '-l-post'],
                  hue='Run Group',
                  size=5,
                  ylim=(0,100), xlim=(0,100),
                  aspect=1)

g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('xx_num-by-area_a3m17_post-steps_both-run-groups.png')

```



Number by area, B1M29, post steps, run group 1

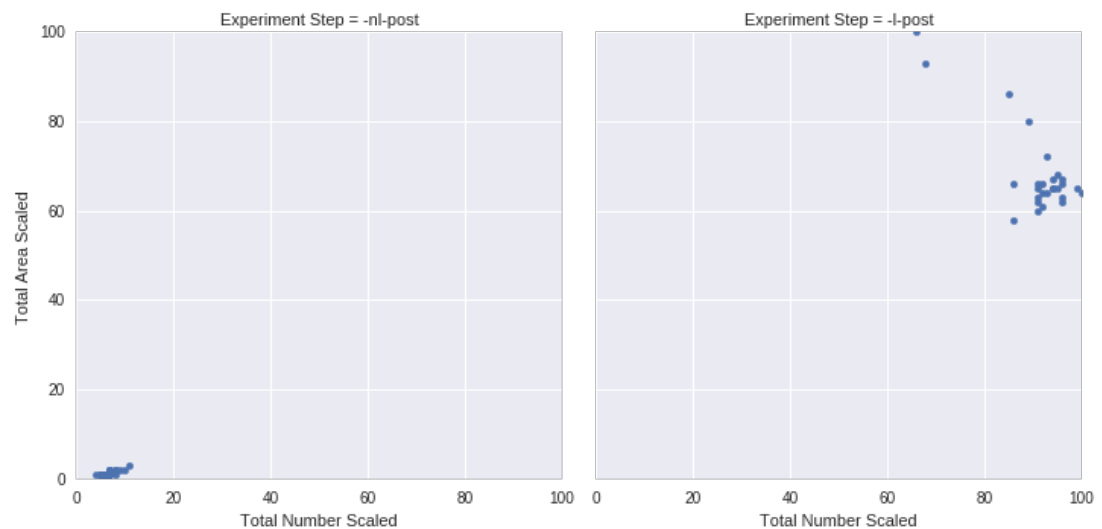
```

In [9]: tmpdf = None
tmpdf = totalsXX.loc[(totalsXX['Receptor'] == 'B1M29') &
                    (totalsXX['Experiment Step'].str.contains('post')) &
                    (totalsXX['Run Group'] == 1)
                    ]

g = sns.FacetGrid(tmpdf,
                  col='Experiment Step',
                  col_order=['-nl-post', '-l-post'],
                  col_wrap=2,
                  size=5,
                  ylim=(0,100), xlim=(0,100),
                  aspect=1)

g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('xx_num-by-area_b1m29_post-steps_run-group-1.png')

```



Number by area, B1M29, post steps, run group 2

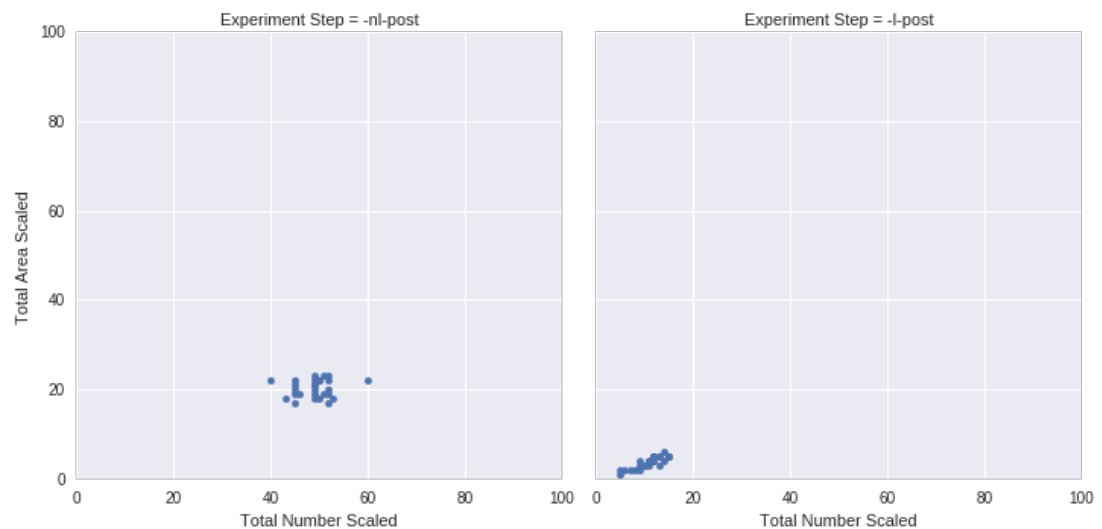

```

In [10]: tmpdf = None
tmpdf = totalsXX.loc[(totalsXX['Receptor'] == 'B1M29') &
                    (totalsXX['Experiment Step'].str.contains('post')) &
                    (totalsXX['Run Group'] == 2)
                    ]

g = sns.FacetGrid(tmpdf,
                  col='Experiment Step',
                  col_order=['-nl-post', '-l-post'],
                  col_wrap=2,
                  size=5,
                  ylim=(0,100), xlim=(0,100),
                  aspect=1)

g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('xx_num-by-area_b1m29_post-steps_run-group-2.png')

```



Number by Area, B1M29, post steps, both run groups

```
In [11]: tmpdf = None
tmpdf = totalsXX.loc[(totalsXX['Receptor'] == 'B1M29') &
                    (totalsXX['Experiment Step'].str.contains('post'))
                    ]

g = sns.FacetGrid(tmpdf,
                  col='Experiment Step',
                  col_order=['-nl-post', '-l-post'],
                  hue='Run Group',
                  size=5,
                  ylim=(0,100), xlim=(0,100),
                  aspect=1)

g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('xx_num-by-area_b1m29_post-steps_both-run-groups.png')
```

